A data structure for monomial ideals with applications to signature Gröbner bases

Théo Ternier
Joint work with Pierre Lairez & Rafael Mohr
MATHEXP, Inria, France

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Processing of signatures (AlgebraicSolving.jl)

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$\mathbf{eco}^{14}$: 14 polynomials in 14 variables

Number of membership tests in a monomial ideal:

117,861,802

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

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If n number of variables and N number of generators of I then:

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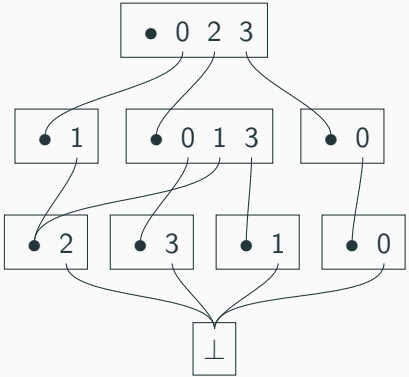
Complexity with our data structure

If n number of variables and N number of generators of I then:

$$O(n\,\ln(N))$$

Monomial divisibility diagram of $I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle$

$m = x^2y^2z^2$ in I?



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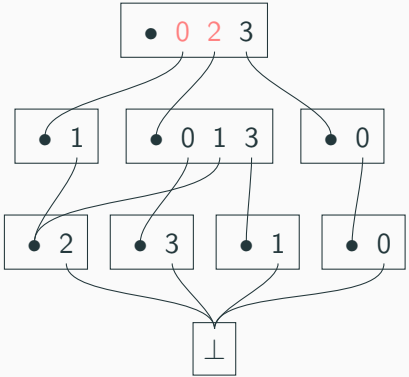

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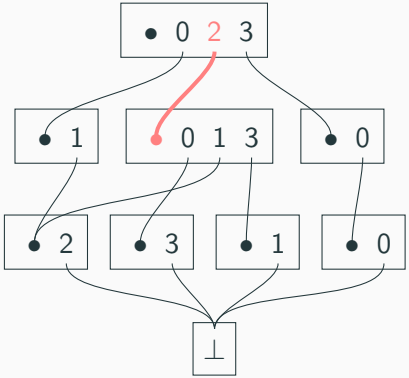

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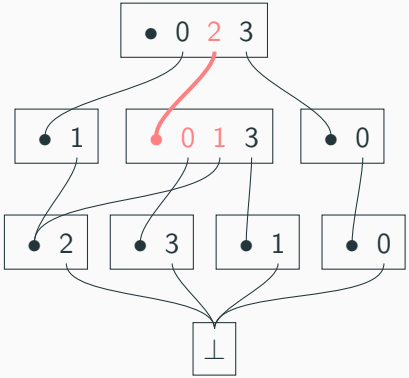

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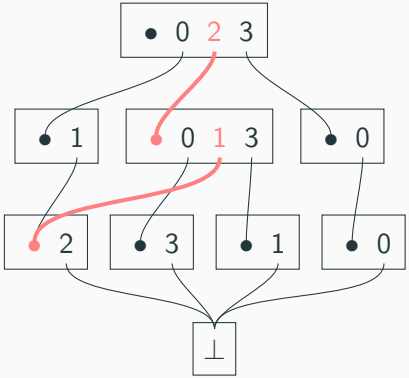

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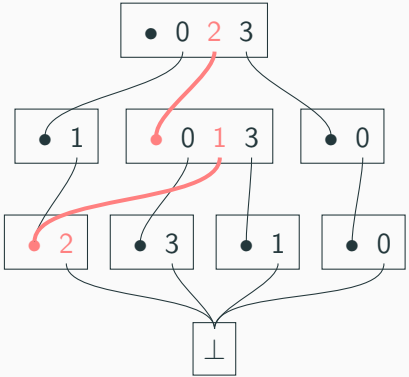

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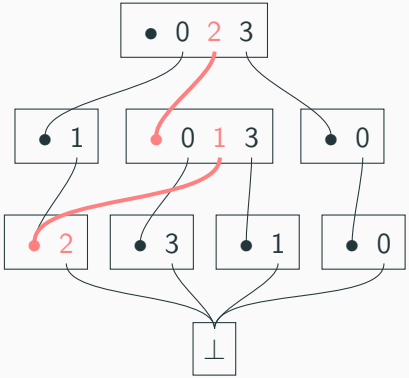

Theorem

For every monomial ideal I, there is a unique minimal directed acyclic graph that supports membership tests in $O(n \ln N)$.

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Example: monomial divisibility diagram of eco^{14}

Number of gen: 2852 Number of nodes: 173 Number of coeffs: 2852 $\times$ 14

Signature Gröbner basis computations

	AlgebraicSolving.jl	*AlgebraicSolving.jl* with mdd					
name	time	time	test	mdd	size	nbr test	×
katsura^{12}	27s	14s	0.17s	0.27s	103	1503655	1.9
katsura^{13}	3min30	1min45	0.71s	1.14s	125	5863729	1.9
cyclic^{7}	1s	0.8s	0.02s	0.02s	43	239572	1.6
cyclic^{8}	1min	29s	0.5s	0.36s	80	4145249	2.1
eco^{14}	3h30	1h06	6s	28s	173	117861802	3.1
noon^{7}	10s	3s	0.08s	0.02s	71	985000	3.1
noon^{8}	3min	44s	0.63s	0.1s	101	7400000	3.5

AlgebraicSolving.jl: Christian Eder (Kaiserslautern), Mohab Safey El Din (Sorbonne Université) and Rafael Mohr (Leuven).